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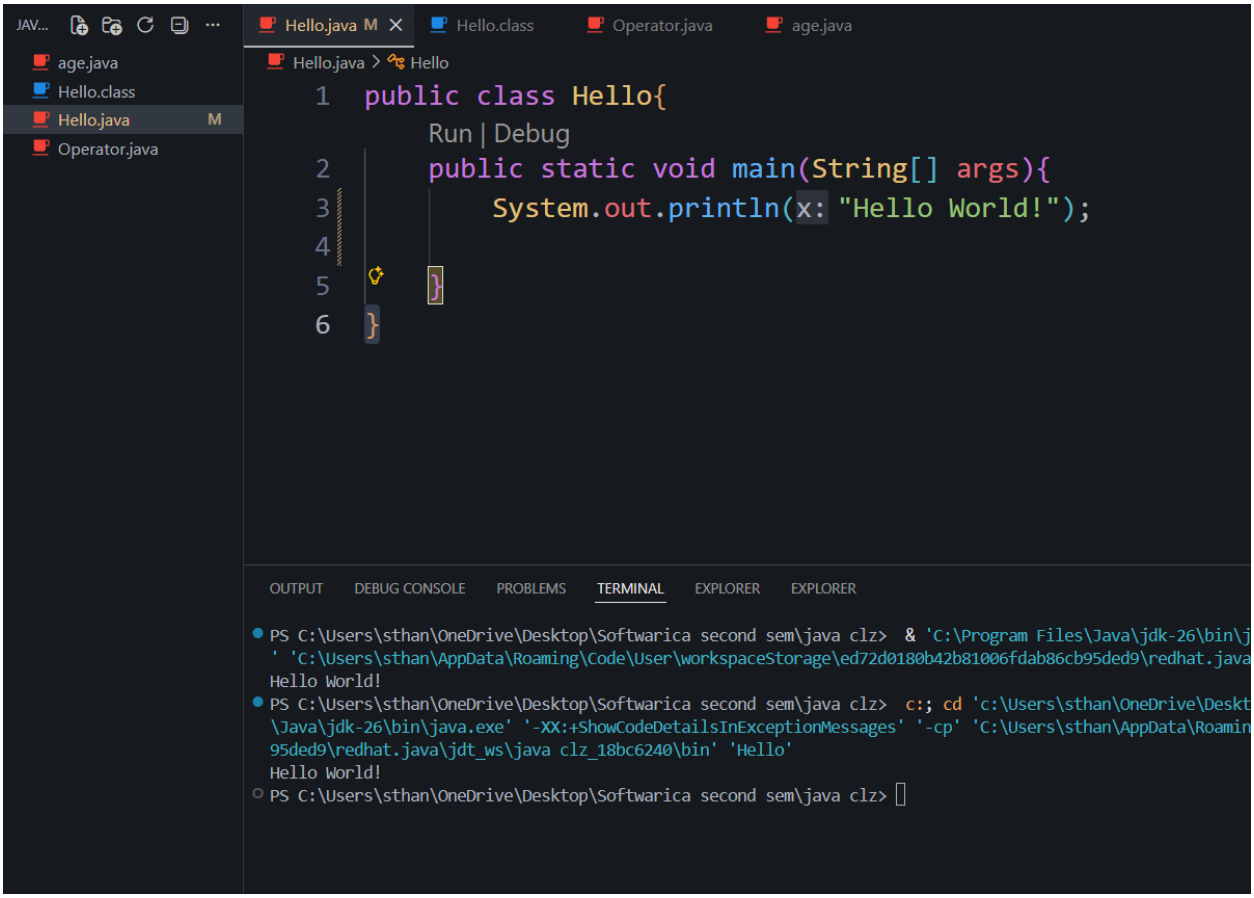
ST4003CEM Object Oriented Programming

Prabisha Khadka

1. Write a program that prints **Hello World!** to the console. Save the program as Hello.java.



2. Extend/Modify the above program to print two additional lines:

   This is my first program

   I am on module ST4003CEM

   System.out.print (**not** println) and produces the same output.

   i. With println. (it have \n)

Hello.java M × Hello.class Operator.java age.java

Hello.java > Hello > main(String[])

```java
public class Hello{
    Run | Debug
    public static void main(String[] args){
        // System.out.println("Hello World!");
        System.out.println(x: "This is my first program");
        System.out.println(x: "I am on module ST4003CEM");

    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX
' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b81006fdab86cb95ded9\redhat.java\jdt_ws\java
Hello World!
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwarica
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\w
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
Hello World!
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwarica
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\w
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
This is my first program
I am on module ST4003CEM
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

ii. With print (it don't have \n)

Hello.java M × Hello.class Operator.java age.java

Hello.java > Hello > main(String[])

```java
public class Hello{
    Run | Debug
    public static void main(String[] args){
        // System.out.println("Hello World!");
        System.out.print(s: "This is my first program");
        System.out.print(s: "I am on module ST4003CEM");

    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-X
' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b81006fdab86cb95ded9\redhat.java\jdt_ws\java
Hello World!
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwaric
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
Hello World!
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwaric
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
This is my first program
I am on module ST4003CEM
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwaric
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
This is my first programI am on module ST4003CEM
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

3. Write a program that prints the following output:

A "quoted" String is

'much' better if you learn

the rules of "escape sequences."

Also, "" represents an empty

String. Don't forget: use \"

instead of " !

" is not the same as "

Hello.java M × Hello.class Operator.java age.java

Hello.java > Hello > main(String[])

```java
public class Hello{
    public static void main(String[] args){
        System.out.println(x: "A \"quoted\" String is");
        System.out.println(x: "'much' better if you learn \nthe rules of \"escape sequences.\"");
        System.out.println(x: "Also, \"\" represents an empty");
        System.out.println(x: "String. Don't forget: use \\\"");
        System.out.println(x: "instead of \"!");
        System.out.println(x: "\"is not the same as \"");


    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz'; & 'C:\Program F
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b81006fdab
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
A "quoted" String is
'much' better if you learn
the rules of "escape sequences."
Also, "" represents an empty
String. Don't forget: use \"
instead of "!
"is not the same as "
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

4. WAP that prints the following pattern:

```
*
**
***
****
*****
```

```java
Hello.java > Hello > main(String[])
public class Hello{
    public static void main(String[] args){
        // System.out.println("\" is not the same as \"");
        System.out.println("*");
        System.out.println("**");
        System.out.println("***");
        System.out.println("****");
        System.out.println("*****");



    }
}
```

5. WAP prints the following pattern:

```
*********
*********
*********
*********
*********
```

Hello.java M × | Hello.class | Operator.java | age.java

Hello.java > Hello > main(String[])

```java
public class Hello{
    public static void main(String[] args){
        // System.out.println("***");
        // System.out.println("****");
        // System.out.println("*****");

        for (int i=0;i<5;i++){
            System.out.println(x: "*********");
        }


    }
}
```

OUTPUT | DEBUG CONSOLE | PROBLEMS | TERMINAL | EXPLORER | EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Use
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Use
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
**
***
****
*****
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Use
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Use
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
*********
*********
*********
*********
*********
```

6. WAP that prints the following output:

A well-formed Java program has a main method with { and } braces.

A System.out.println statement has (and) and usually a String that starts and ends with a " character.

(But we type \" instead!)

What is the difference between

a ' and a "? Or between a " and a \"?

One is what we see when we're typing our program. The other is what appears on the "console."

```
Hello.java M   Hello.class   Operator.java   age.java
Hello.java > Hello > main(String[])
public class Hello{
    public static void main(String[] args){
        System.out.println(x: "A well-formed Java program has a main method with { and } braces. \n");
        System.out.println(x: "A System.out.println statement has ( and ) and usually a String that");
        System.out.println(x: "starts and ends with a \" character.");
        System.out.println(x: "(But we type \\\" instead!)");
        System.out.println(x: "What is the difference between");
        System.out.println(x: "a ' and a \"?  Or between a \" and a \\\"? \n");
        System.out.println(x: "One is what we see when we're typing our program. The other is what appears on \nthe \"console.\"");
    }
}
```

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\User
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\User
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'Hello'
A well-formed Java program has a main method with { and } braces.

A System.out.println statement has ( and ) and usually a String that
starts and ends with a " character.
(But we type \" instead!)
What is the difference between
a ' and a "?  Or between a " and a \"?

One is what we see when we're typing our program. The other is what appears on
the "console."
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```